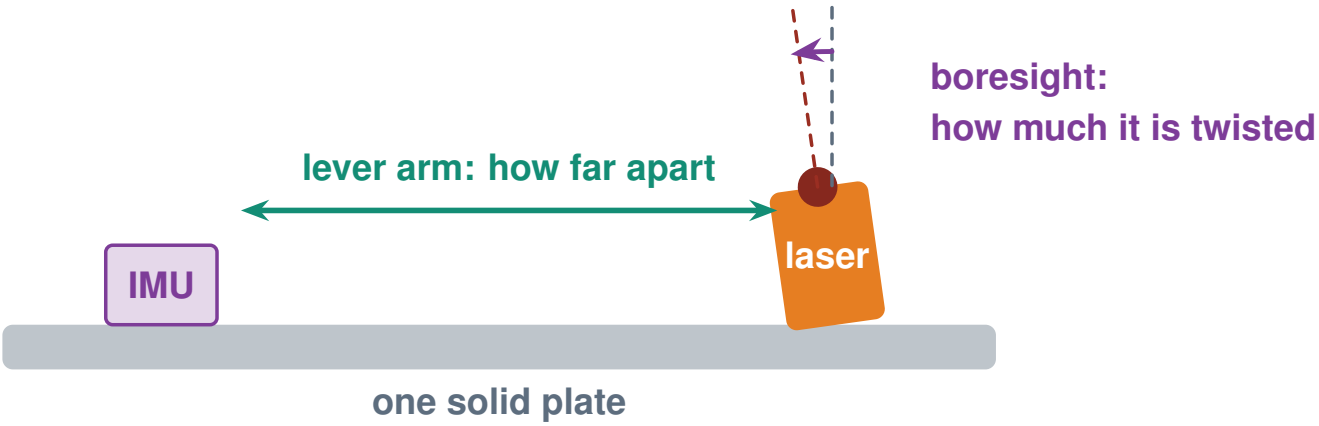
Lever Arm and Boresight Calibration

BIG IDEA

We must measure exactly how far apart, and how twisted, the sensors are.



THE IDEA

Put a toy camera on your shoulder and close your eyes. You know where *you* are, but the camera is a little to the side, a little higher, and maybe turned a bit. Forget that, and you point to the wrong place. On a mapping car the motion sensor (the IMU) and the laser are in different spots too.

REMEMBER

- ✓ The **lever arm** is how far apart the sensors are.
- ✓ The **boresight** is how much one is twisted compared with the other.
- ✓ Check them again after the mount is bumped, moved or stored for a long time.

TRY THIS

Tape a straw to the side of a box so it points a little off to one side. Aim the box at a target. Where does the straw really point?

The full module is in the Reference Book.